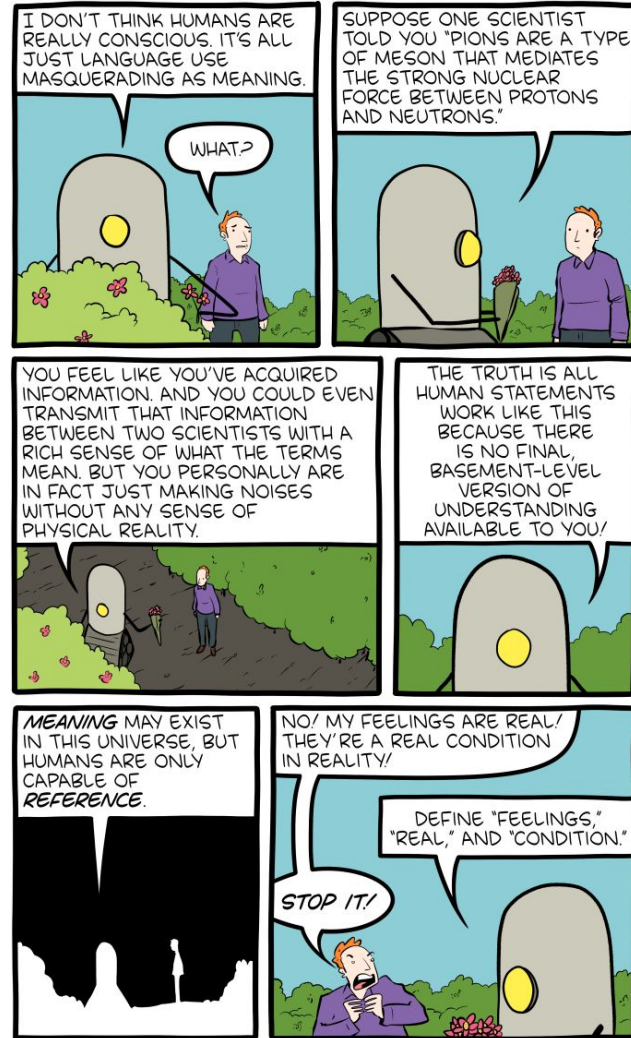


CS 159 Week 8: Wrapup - LLMs, and the future (and past) of NLP



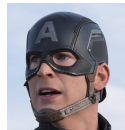
(<https://www.smbc-comics.com/comic/consciousness-7>)

Recap: Previously on CS 159...

- Week 3: Language models
- Week 4: Text classification
- Week 5: Hidden Markov models
- Week 6: Information retrieval
- Week 7: Vector semantics

Assembling the NLP super-team

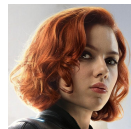
- Week 3: Language models



- Week 4: Text classification



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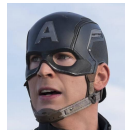


- Week 7: Vector semantics



Assembling the NLP super-team

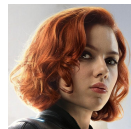
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- Week 7: Vector semantics



Motivation: Building better language models



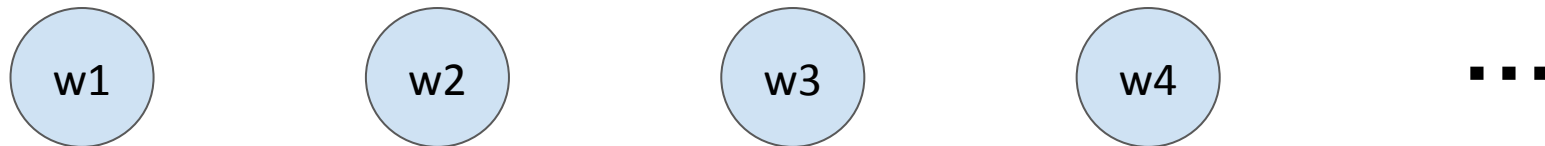
$$P(w_1, w_2, w_3, \dots, w_k) \approx ???$$

Starting point: bag-of-words (unigram) LM

$$P(w_1, w_2, w_3, \dots, w_k) \approx \prod P(w_i)$$

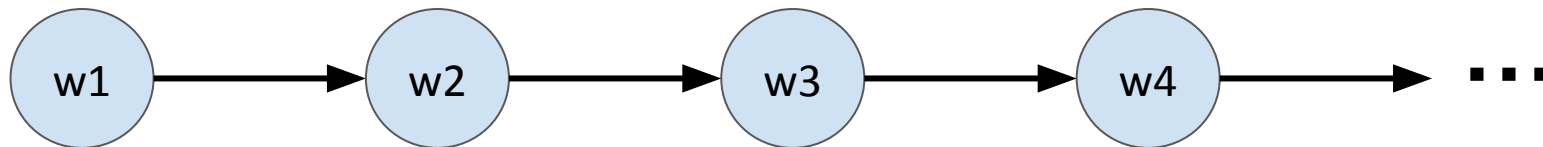
Starting point: bag-of-words (unigram) LM

$$P(w_1, w_2, w_3, \dots, w_k) \approx \prod P(w_i) \quad (\text{context size} = 0)$$



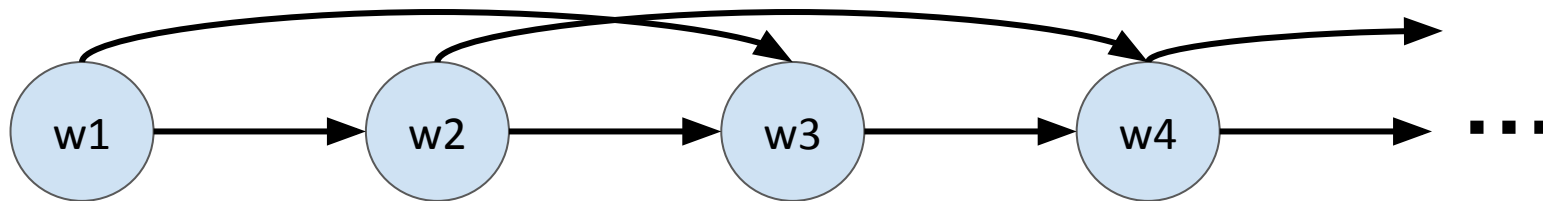
Improvement: bigram Markov LM

$$P(w_1, w_2, w_3, \dots, w_k) \approx \prod P(w_i | w_{i-1}) \quad (\text{context size} = 1)$$



Further improvement: trigram Markov LM

$$P(w_1, w_2, w_3, \dots, w_k) \approx \prod P(w_i | w_{i-2}, w_{i-1}) \quad (\text{context size} = 2)$$



More context is better, but...

In theory, the bigger the context window, the more realistic the text generated by our **language model**



In practice, actually training a generalizable large-context Markov LM would require essentially infinite data!

$P(\text{"Pomona"} | \text{"Board", "bus", "197", "from", "the", "terminal", "in"})$



(need to compute)

$C(\text{"Board", "bus", "197", "from", "the", "terminal", "in", "Pomona"})$

How many times would you see this exact sequence in the data?

More context is better, but...

In theory, the bigger the context window, the more realistic the text generated by our **language model**



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$P(\text{"Pomona"} | \text{"Catch", "the", "bus", "at", "the", "station", "in"})$



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What about this one?

Solving the context sparsity problem

Previously, we have seen naive solutions such as add-k smoothing, which assume that previously-unseen contexts have some *constant* (low) probability

But we can be smarter about this if we realize that “Board bus 197 from the terminal in” means roughly the same thing as “Catch the bus at the station in”!

We need a model of *semantic* similarity...

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We need a model of *semantic* similarity... **vector semantics** !



Building a language model



with vector semantics



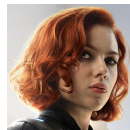
Building a language model



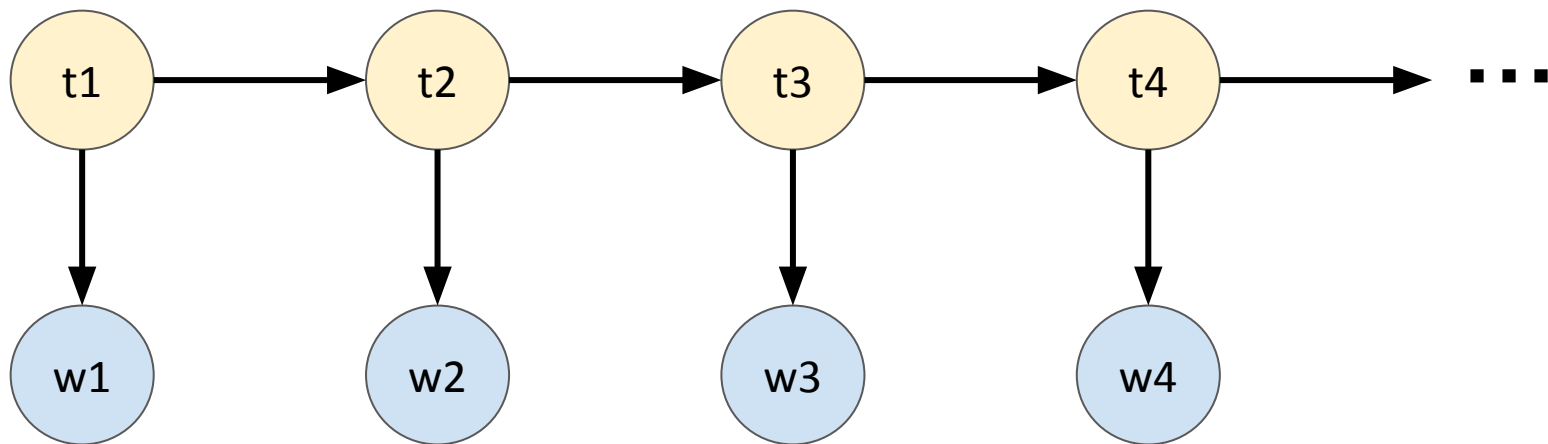
with vector semantics



Remember Hidden Markov models



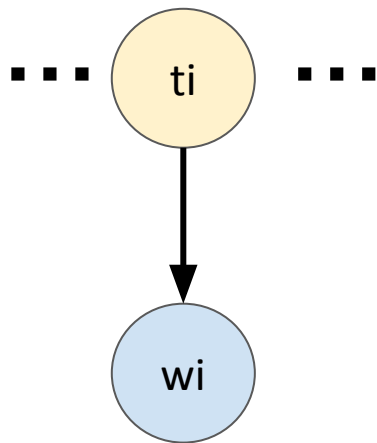
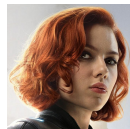
?



Building a **language model** with **vector semantics**



Remember **Hidden Markov models** ?



Hidden states (parts of speech)

produce

Observed words

Building a **language model**

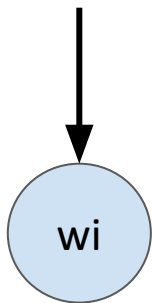


with **vector semantics**



Hidden **vectors**

produce



Observed **words**

Building a **language model**



with **vector semantics**



Consider a Markov model with a *context window* of size c , such that the probability of the next word w_i is computed as:

$$P(w_i | w_{i-1}, w_{i-2}, \dots, w_{i-c})$$

Let context be a sequence of *word vectors*, one for each word in the context window: $\text{context} = [v_{i-1}, v_{i-2}, \dots, v_{i-c}]$

Then we will compute the above probability as some function f of context: $P(w_i | w_{i-1}, w_{i-2}, \dots, w_{i-c}) = f(\text{context})$

A problem with the proposed model

$f(\text{context})$ is not a straightforward thing to interpret.

We defined context as a *sequence* of vectors—what does it mean to have a function on a *sequence* of vectors?

On the other hand, if we had a *single* vector representing the entire context, we could imagine functions on that vector (think back to linear algebra!)

Turning a sequence of vectors into a single vector

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Naive idea: take a sum or average of the vectors

Problem: not all words are equally important...and *which* words are important depends on the context itself!

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A better idea: take a *weighted* sum of the vectors, where words are weighted according to how important they are

Problem: how do we find the important words?

Turning a sequence of vectors into a single vector

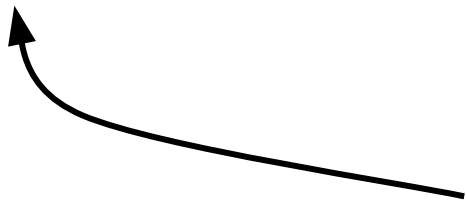
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Turning a sequence of vectors into a single vector

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Problem: how do we **find** the important words?



This sounds like **information retrieval**



!

Pay attention to Attention!

For *every* word w_i in the context, represented as a word vector x_i , we compute:

- A **query vector** $q_i = x_i W^Q$
- A **key vector** $k_i = x_i W^K$
- A **value vector** $v_i = x_i W^V$

Where W^Q , W^K , W^V are matrices optimized using a neural network

Pay attention to Attention!

Given a context whose last word is w_i :

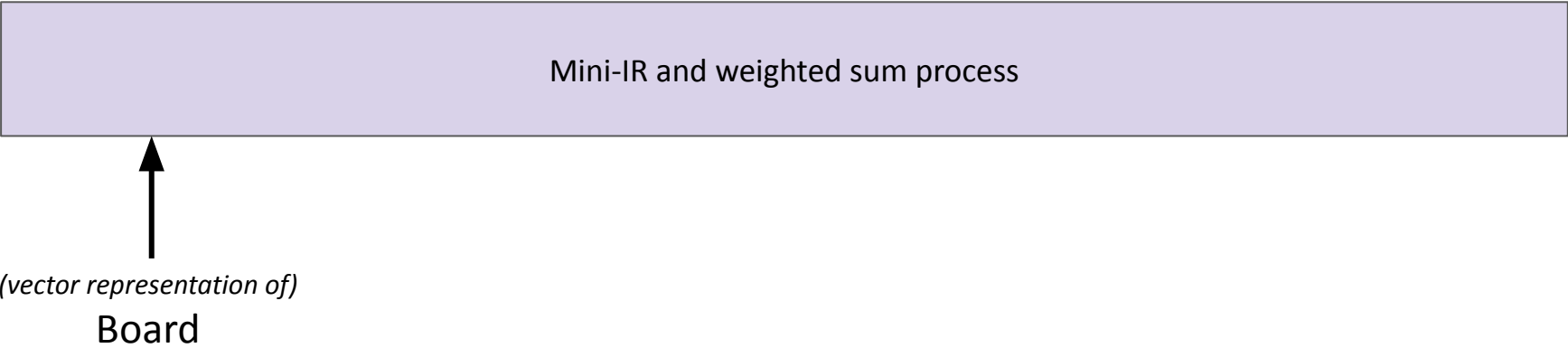
For *each* word w_j in the context, we compute an **attention weight** α_{ij} using a “mini-IR” process:

1. Let $\text{score}(i,j) = \text{cosine}(q_i, k_j)$
2. Normalize $\text{score}(i,j)$ across all j , and save the normalized $\text{score}(i,j)$ as α_{ij}

Then, we compute a vector representation of the entire context as the weighted sum $\sum_j \alpha_{ij} v_j$

(vector representation of)

Board



Mini-IR and weighted sum process

The diagram consists of a light purple rectangular box with a thin black border. Inside the box, the text "Mini-IR and weighted sum process" is centered. Below the box, on the left side, is a black arrow pointing upwards. To the left of the arrow, the text "(vector representation of)" is written in italics, and below that, the word "Board" is written in a standard font.



(vector representation of)

Board

(new vector)

x_1^0



Mini-IR and weighted sum process

(vector representation of)

Board



(new vector)

x_1^0



Mini-IR and weighted sum process

(vector representation of)

Board

(vector representation of)

bus



(new vector)

x_1^0



Mini-IR and weighted sum process



(vector representation of)

Board

(vector representation of)

bus

(new vector)

x_1^0



(new vector)

x_2^0



Mini-IR and weighted sum process

(vector representation of)

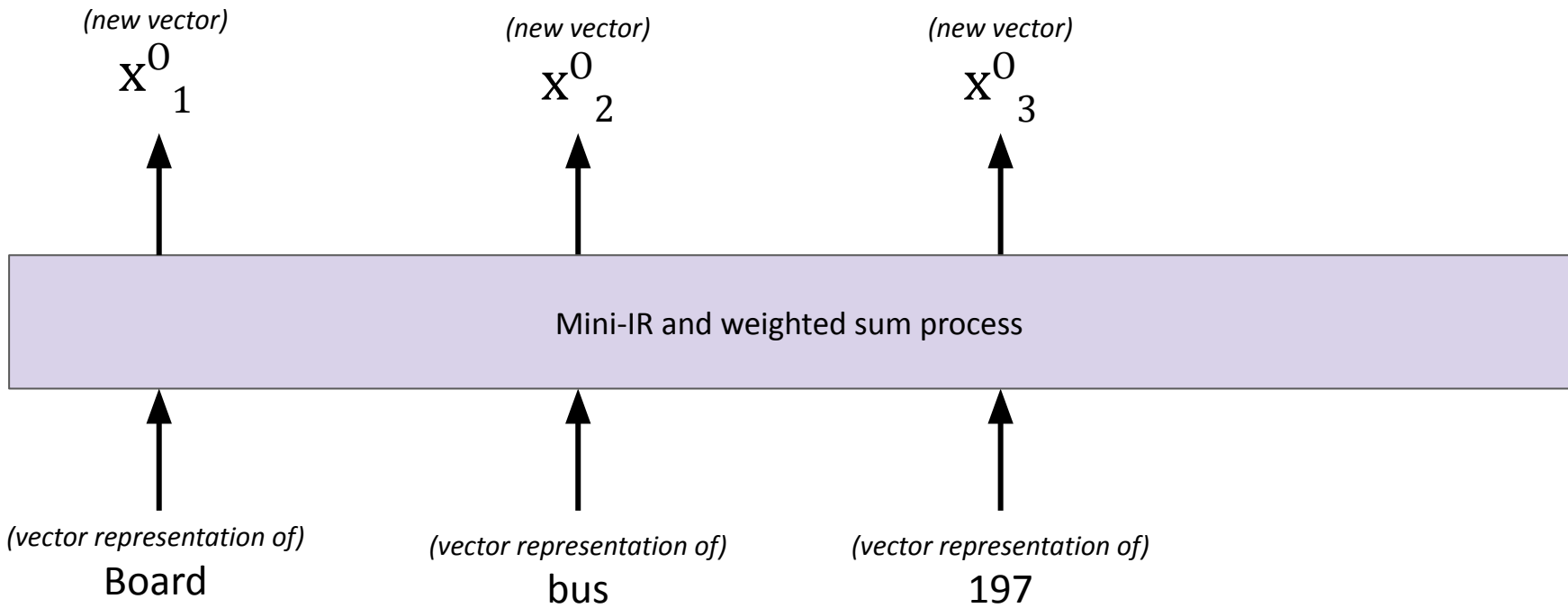
Board

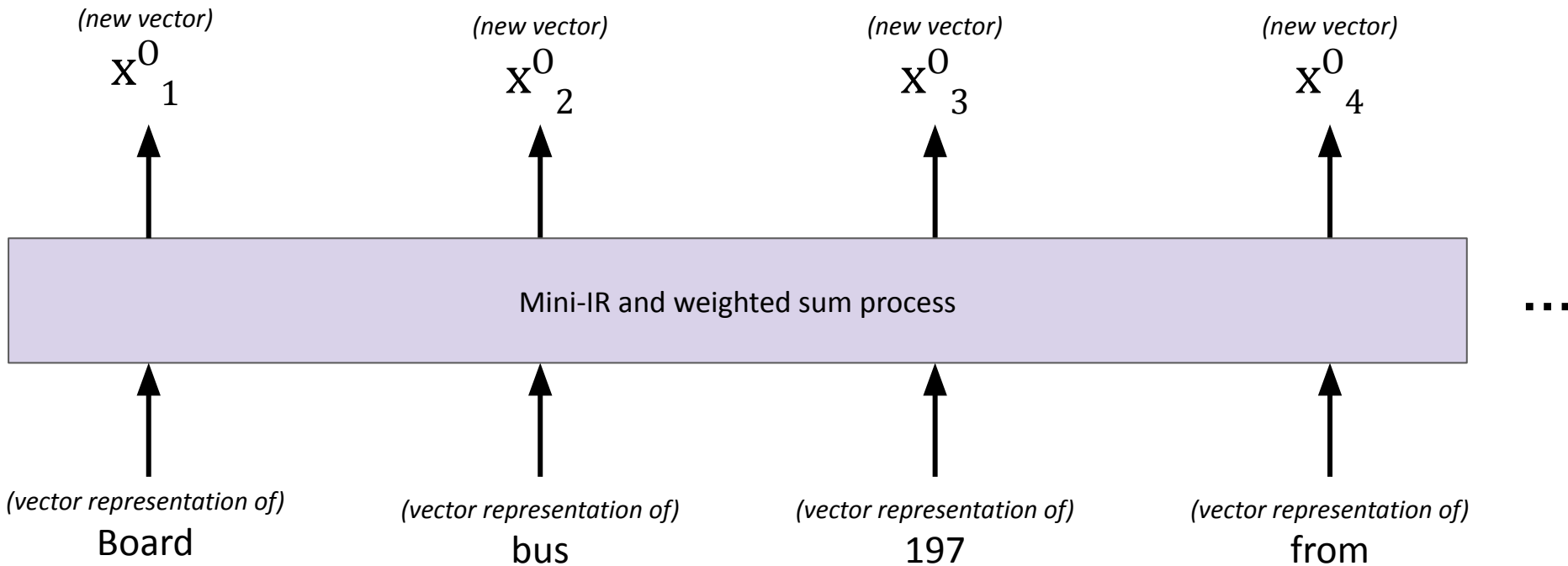


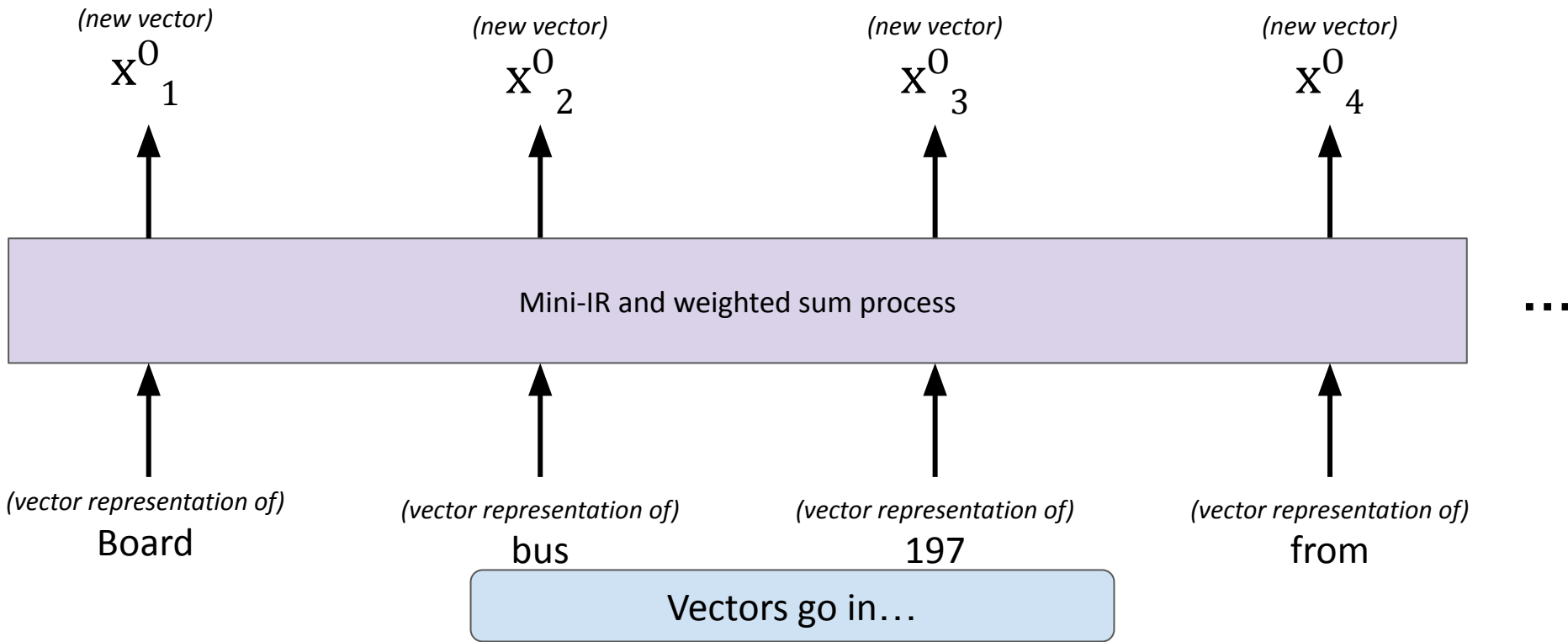
(vector representation of)

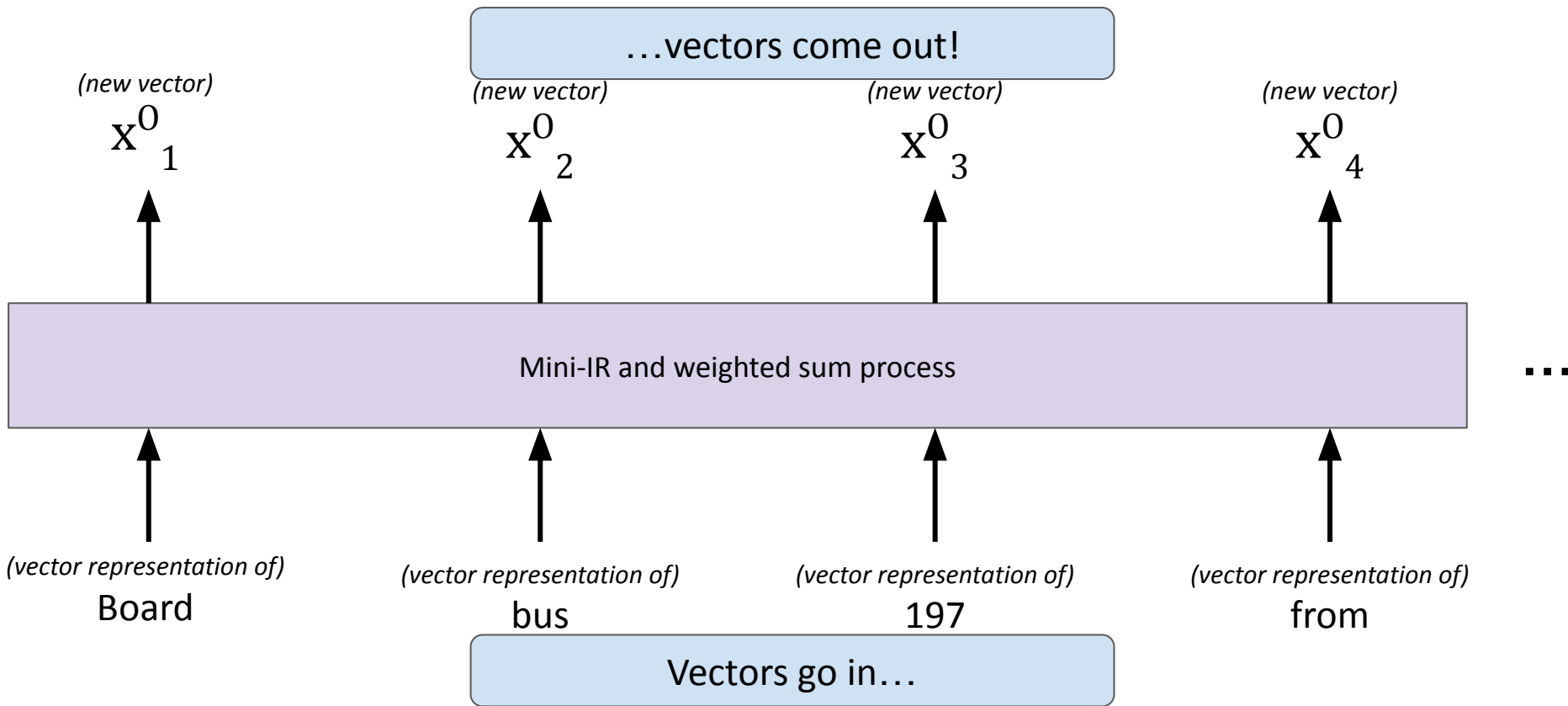
bus



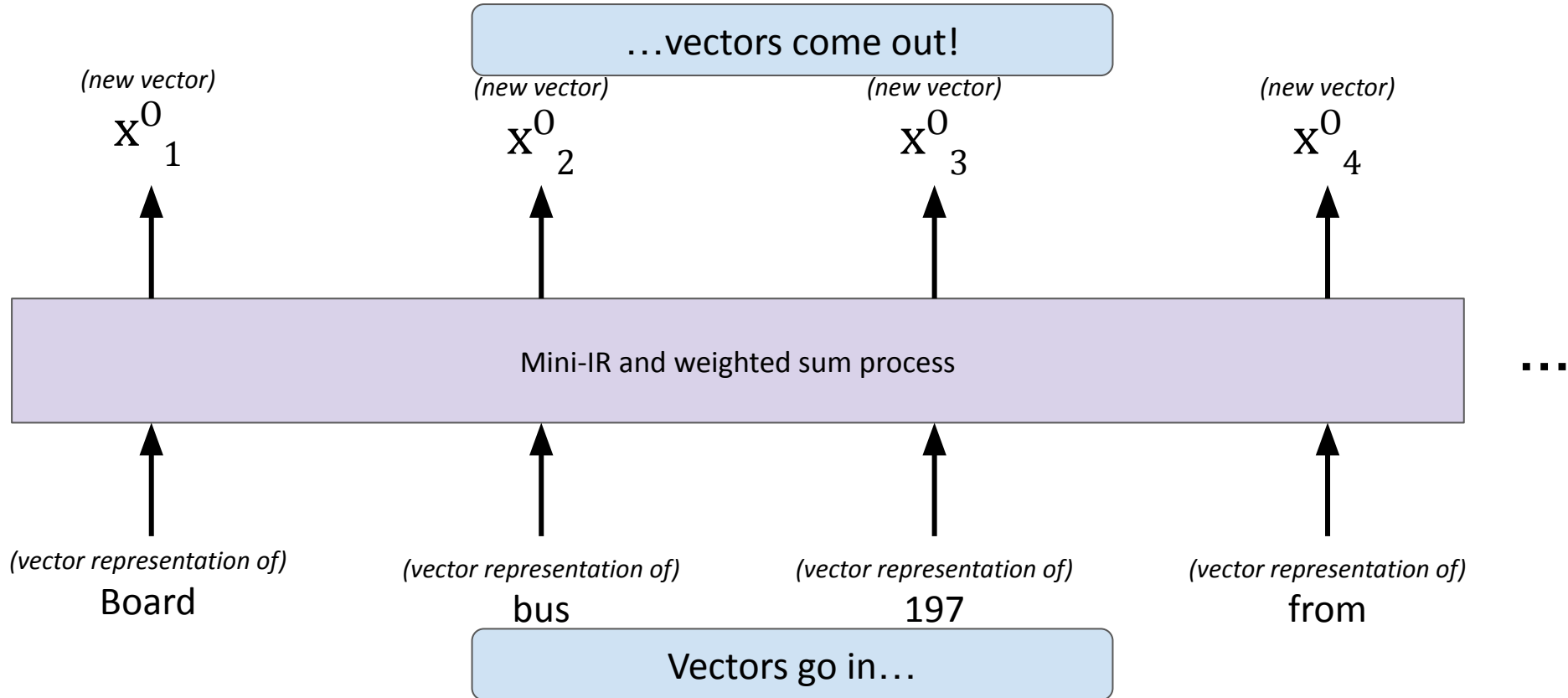




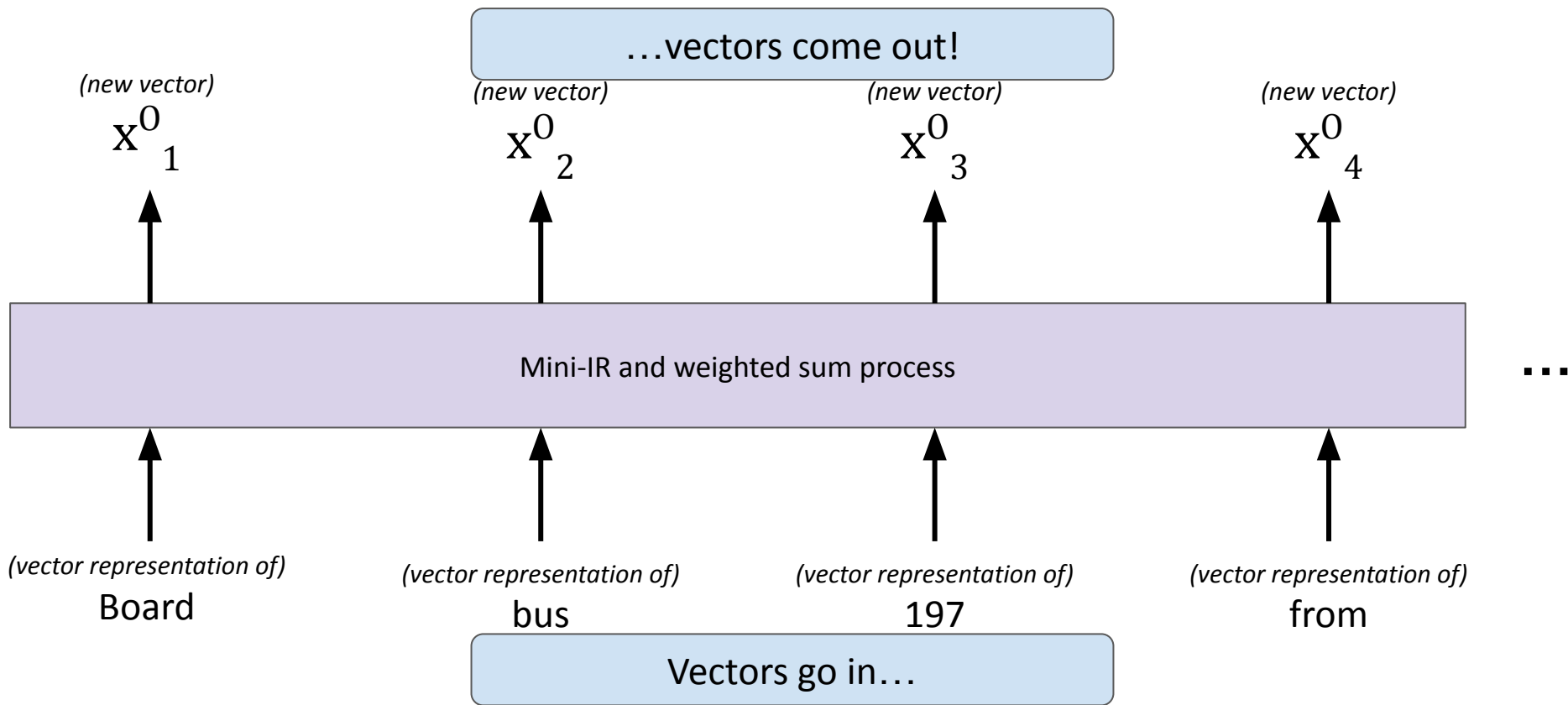




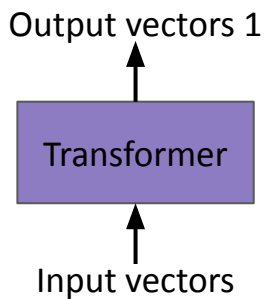
This process *transforms* vectors from one space to another



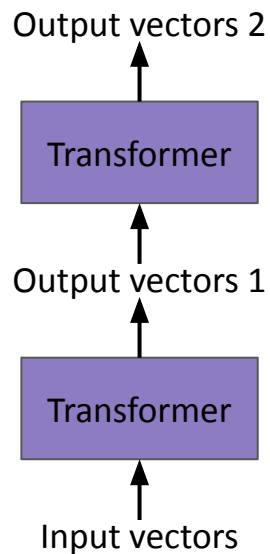
The **transformer** *transforms* vectors from one space to another



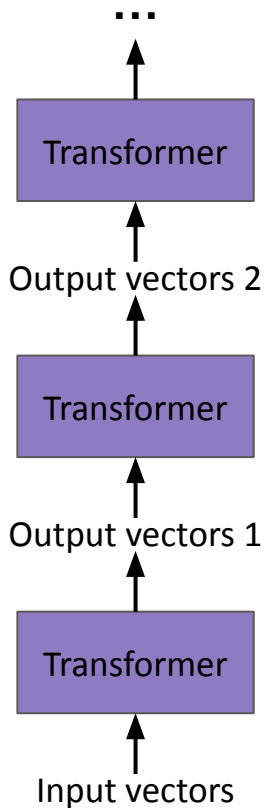
What's better than a transformer? Many transformers!



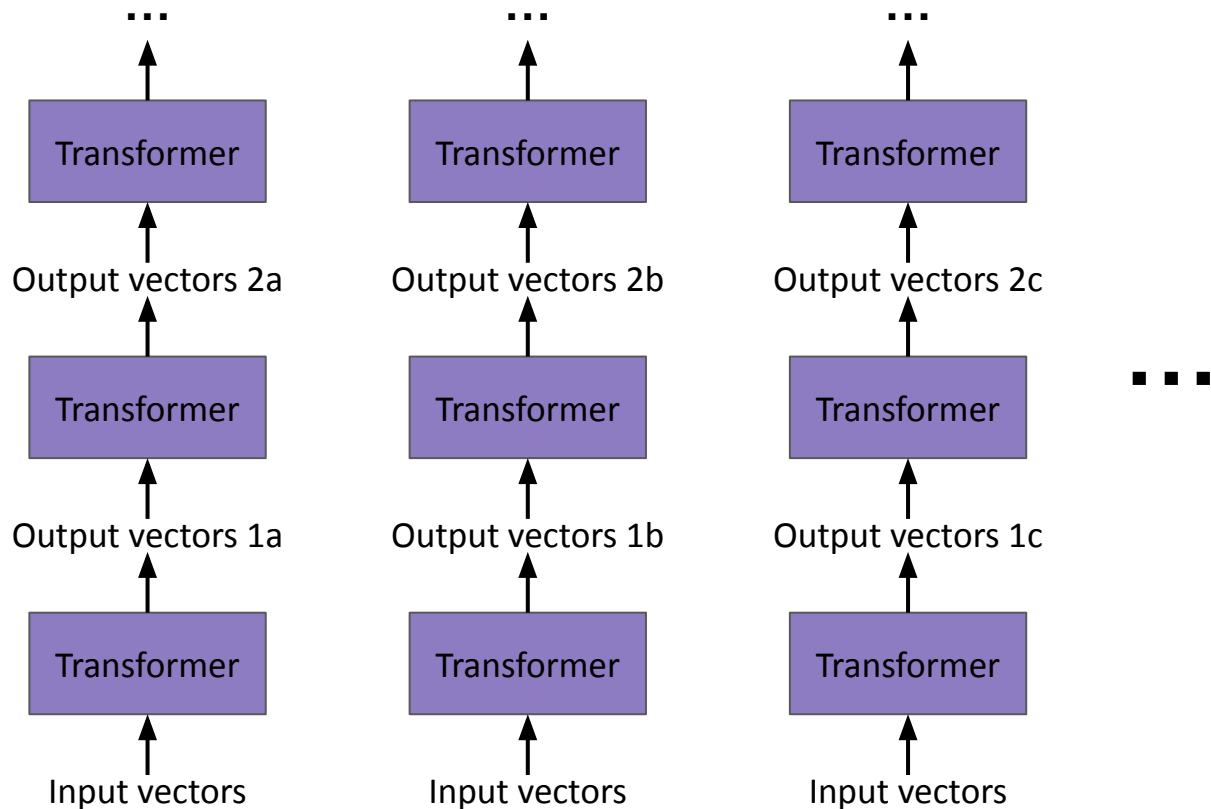
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What's better than a transformer? Many transformers!



Building a language model



with transformers

We've figured out how to turn context into a single vector...now the last step is computing $P(w_i | w_{i-1}, \dots, w_{i-c}) = f(\text{context})$

What is f ?

Building a language model



with transformers

We've figured out how to turn context into a single vector...now the last step is computing $P(w_i | w_{i-1}, \dots, w_{i-c}) = f(\text{context})$

What is f ?

Answer: a text classifier



words in the vocabulary!

where the “classes” are all the

Building a **language model**



with transformers

Refresher: a **text classifier**



models a probability distribution over classes, conditioned on a document:

$$P(c \in C \mid d)$$

In this case, the classes are words, the set C of possible classes is the vocabulary V , and the document is the context:

$$P(w \in V \mid \text{context})$$

We solve the classification task using a neural network.

LLMs and the future of NLP

This was only a light intro to LLMs

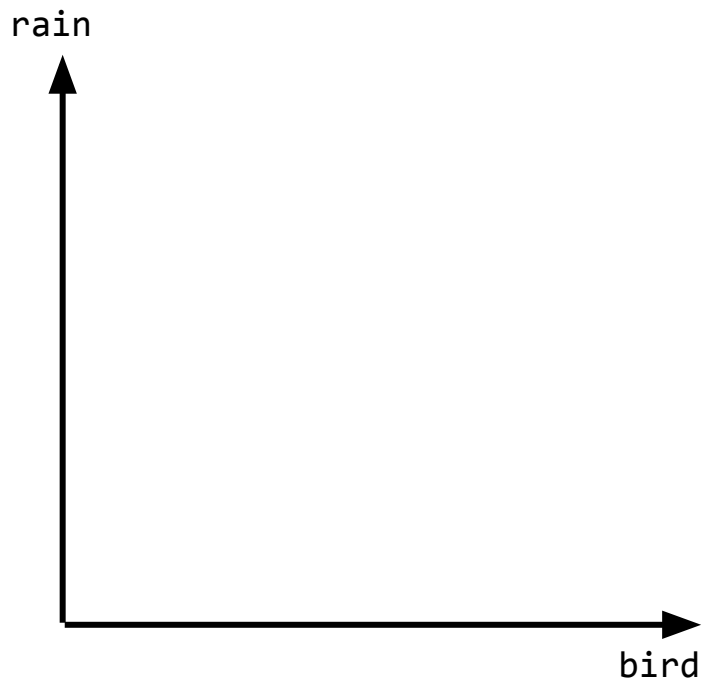
There's much more to explore! (See textbook Chapter 8 if you're interested; this lecture only partly covered 8.1-8.5)

But for now I want to focus on a specific question...

What is the relationship between LLMs and search (information retrieval)?

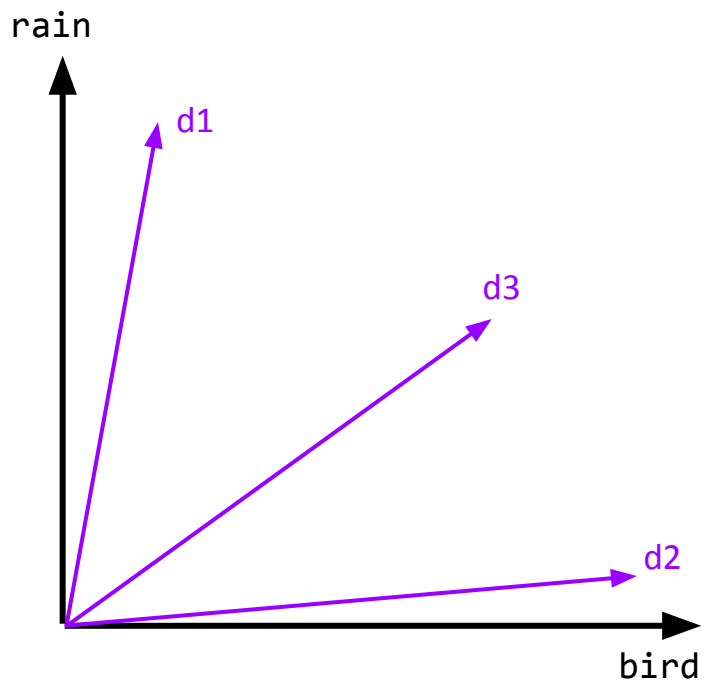
LLMs vs IR: a geometric view

Traditional IR



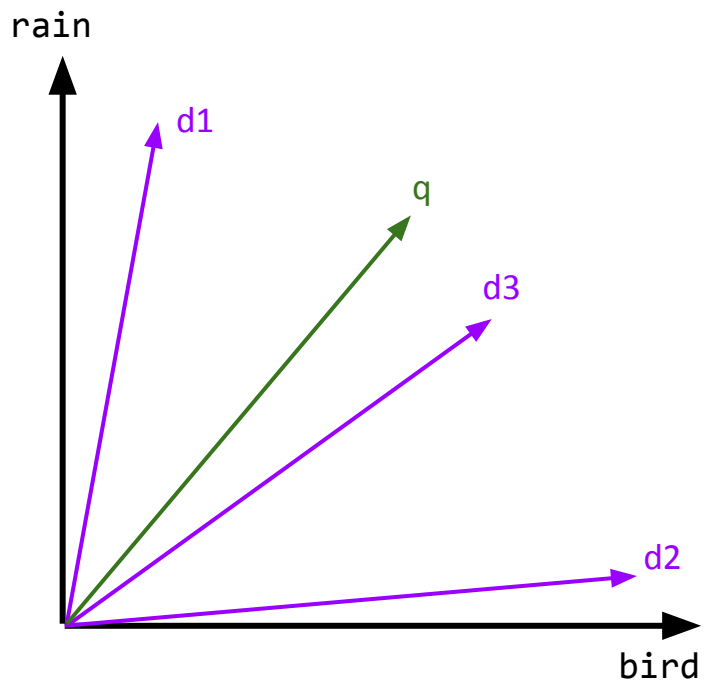
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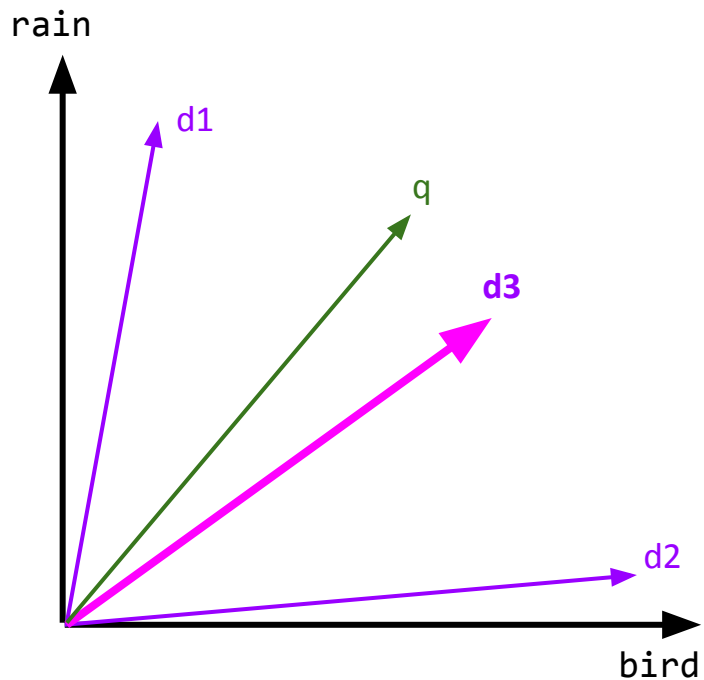
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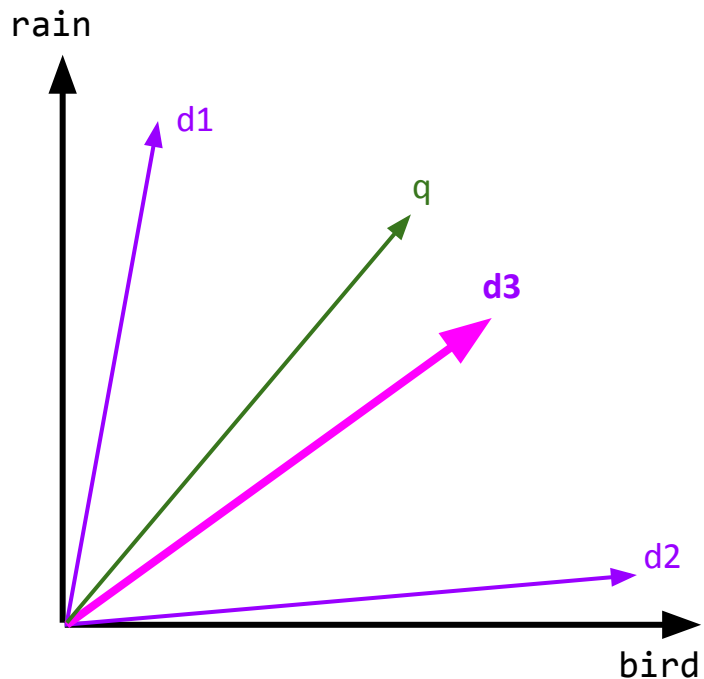
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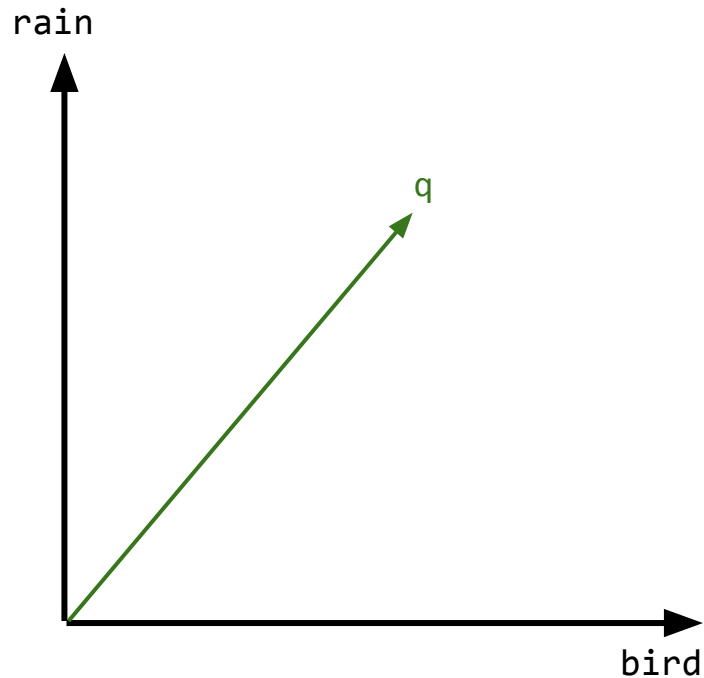


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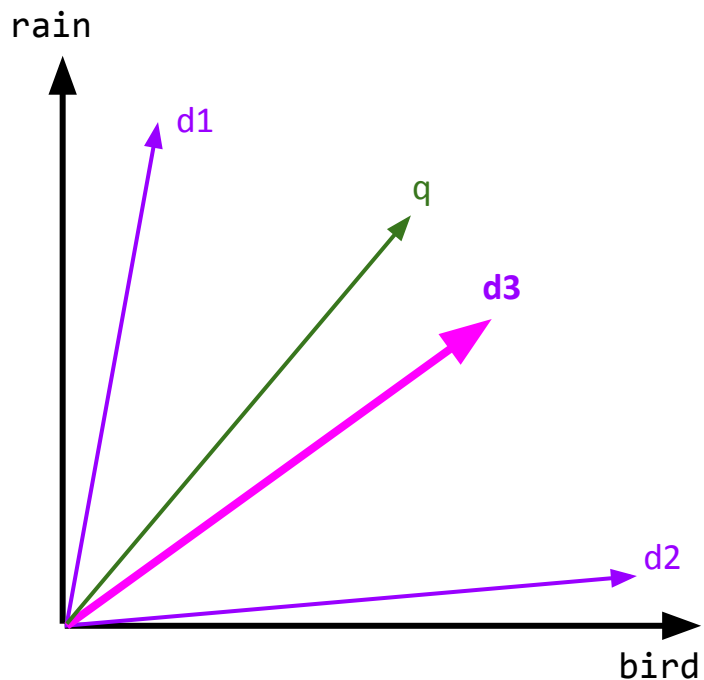


LLM text generation

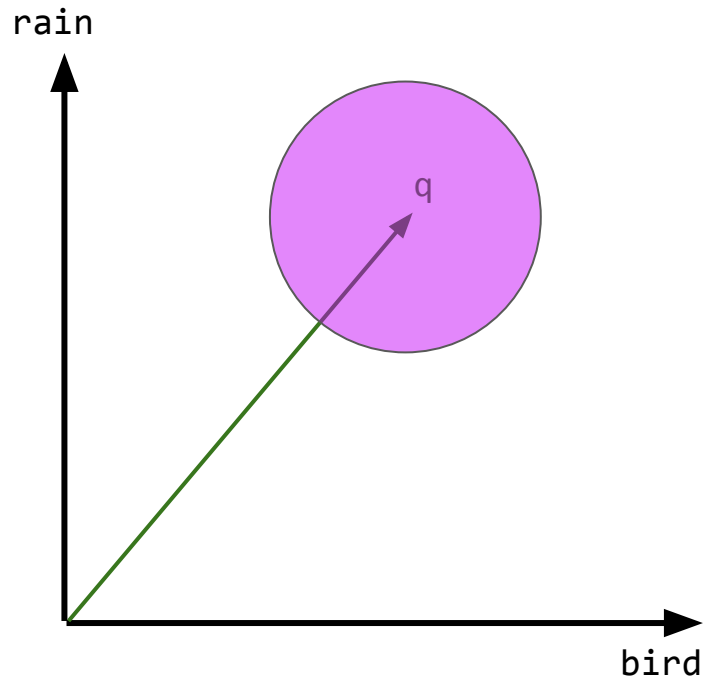


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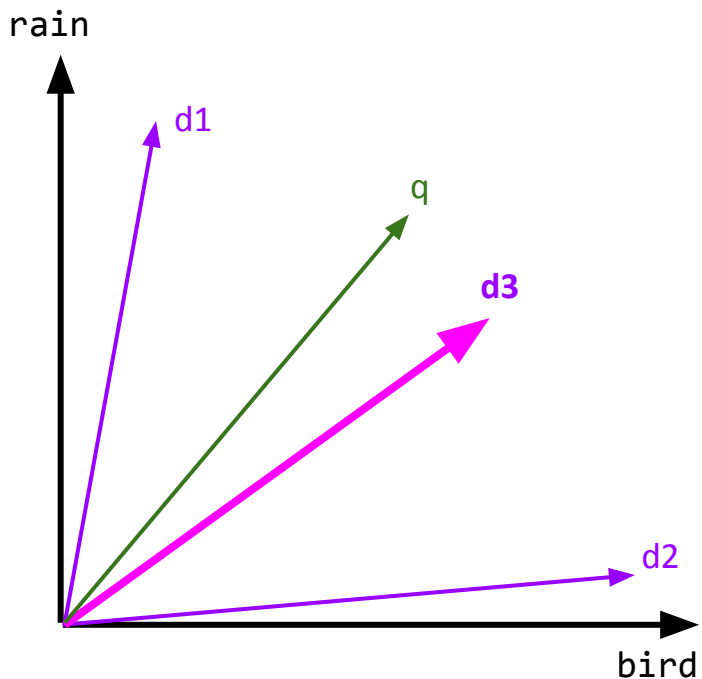


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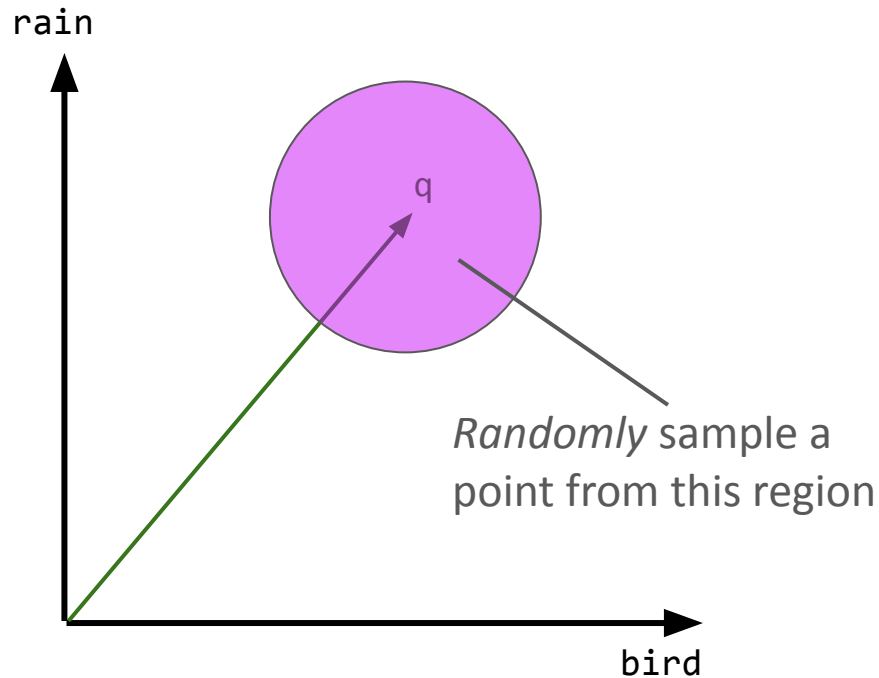


LLMs vs IR: a geometric view

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LLM text generation



LLMs and the future of NLP

I tried using one of the fancy new AI tools (Gemini, ChatGPT Pro, etc.) and it's not making the common mistakes with IR, math, etc. that LLMs make!



LLMs and the future of NLP

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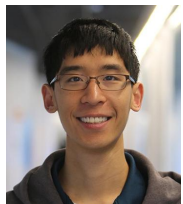
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(stands up on a soapbox that came out of nowhere) And that's the right thing to do! Because, as we've seen, LLMs are good at text generation and bad at other things...but the things LLMs are bad at are often things where we have *other* tools that are *good* at those things. LLMs don't make other approaches obsolete, they all complement each other. And that's something every NLP researcher should keep in mind; always pick the right tool for the job! :)

A journey from **words** to **language** to **meaning**

We've got this!
(Weeks 1-2)

A journey from **words** to **language** to **meaning**

*(but there's still more
that can be done)*

We're feeling pretty good about this too!
(Weeks 3-7)

A journey from **words** to **language** to **meaning**

(but it's still very much a work in progress)

This one's harder...but as we just saw, progress is being made.

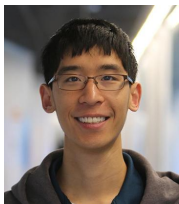
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(here clearly there's a LOT more to do!)

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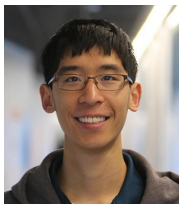


This isn't the end! For the rest of the semester you'll be looking at stuff that NLP researchers have been doing in this direction very recently...

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This isn't the end! For the rest of the semester you'll be looking at stuff that NLP researchers have been doing in this direction very recently...

And perhaps the *next* great breakthrough will be made by none other than you ;)